
Magnificent Seven companies: analysis of revenue trends during COVID pandemic

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Abstract: This research focuses on analysing revenue trends of the Magnificent Seven (M7) companies during the COVID pandemic (2020–2023). Using time series ensemble models, we forecasted quarterly revenues and applied counterfactual analysis to evaluate revenue trends during the pandemic. We determined that the majority of the M7 companies were able to sustain and increase their revenues versus estimated ensemble forecasts during the pandemic years of 2020–2023. We also concluded that for a short-term period, e.g., within the first pandemic year of 2020, some of the M7 companies had experienced revenue losses that were potentially associated with the adverse effects of the pandemic. However, for the subsequent COVID years of 2021–2023, revenue trends for all M7 companies were not adversely affected by the pandemic. This research also provided specific directions on how to develop a new or improve existing methodology for analysing revenues of any organisation during the pandemic.

Keywords: analysis of revenue trends; Magnificent Seven companies; COVID pandemic; time series ensemble models; counterfactual analysis.

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1 Introduction

The COVID pandemic caused significant economic losses for countries worldwide. The drop of US GDP due to the pandemic reached an estimated \$14 trillion in 2020–2023 (Hlavka and Rose, 2023). An extensive number of US industries lost a significant amount of revenue in that period. The US industries that lost the largest proportion of revenue during the first 30 months of the pandemic (from March 2020 to June 2022) were air travel with approximately 58% of decline, dining with close to 66% of decrease, in-store shopping with around 43% of loss, and recreation at almost 40% of decrease (Walmsley et al., 2023).

Not all industries in the US suffered as much as the specified air travel, dining, in-store shopping, and recreation industries. In fact, some of the industries continued to grow revenue during the pandemic above the pre-pandemic levels. However, the existing literature sources and extant research are inconclusive on the revenue trend, i.e., increase, loss, or no change, during the COVID pandemic. Some researchers concluded that large companies, including those in the technology industry, have a higher negative impact in shareholder values associated with their revenue losses (Klößner et al., 2023). On the other hand, several research papers, for example, Hossain et al. (2023), indicated that the high-tech industry benefited from the pandemic, as the increased dependency on this sector during the COVID outbreak resulted in higher revenue for that industry.

In our research, we considered the revenue data of the US organisations that are known as Magnificent Seven (M7) companies. They include Alphabet (the parent company of Google), Amazon, Apple, Nvidia, Meta (Facebook), Microsoft, and Tesla. The Magnificent Seven term was introduced to those seven companies in May 2023 to designate a similar strong revenue growth and impressive stock market performance of each company in 2023 and prior years (Mitra, 2024). The selection of revenue as a measure for analysing the performance of the M7 companies during the COVID pandemic is based on its crucial role in these businesses. Revenue represents the total income generated from the sales of goods or services before any expenses are deducted. Monitoring revenue is essential for assessing a company's financial health, growth potential, and overall performance. It is also a key factor in profitability, cash flow, and valuation.

Most of the M7 companies, with the exception of Amazon and Tesla, belong to the high-tech industry. Amazon operates as both a retail company, through its e-commerce platform and physical stores, and a major player in high technology with its cloud computing services, Amazon web services, artificial intelligence, and other technologies. Similarly, Tesla, known for electric vehicle production, is recognised for its advanced battery technology, self-driving software, and renewable energy innovations. Therefore, both Amazon and Tesla can be considered high-tech companies, at least in part.

The discussed literature approaches that analysed revenue trends throughout the COVID pandemic were predominantly concerned with estimating the revenue loss or, sometimes, revenue increase during the pandemic (Kim et al., 2020; Nair, 2021; Palazzo, 2023). Our literature search did not, however, locate a research publication that would consider revenue trends and their comparison for all M7 companies throughout the pandemic time. These factors motivated our research, which aims to identify and analyse the revenue trends of the M7 companies during the COVID pandemic, as well as assess whether their revenues were adversely affected by the pandemic.

This research produced several important contributions to the theory and practice of analysing revenue trends throughout the COVID onslaught. The first important contribution involves conducting a novel study which explored the effect of the COVID pandemic on revenue trends of the M7 companies. Another contribution is the innovative utilisation of the time series ensemble model to forecasting quarterly revenue values during the COVID pandemic. One more contribution includes applying counterfactual analysis to identify revenue trends in each M7 company throughout the pandemic. Finally, we made important conclusions regarding the adverse effect (or no effect) of the COVID pandemic on revenue trends in the M7 companies. Based on these results, our research provided specific directions on how to develop a new methodology and practical application of analysing revenue trends (revenue loss, increase, or no change) for any organisation during the COVID pandemic.

This paper's structure includes five main sections. After the introduction, the literature review section analyses extant research pertaining to revenue trend in high-tech companies and, specifically, in the M7 companies during the pandemic. The methodology section provides a description of the research objectives, data collection, and methods used in this research. The analysis, results, and discussion section presents analytical results and discussion of revenue trends in the M7 companies during the COVID pandemic using the designed ensemble model of time series forecasting and counterfactual analysis. The conclusion section provides a summary of this research, its results and limitations.

2 Literature review

The topics of revenue evaluation, analysis, and prediction have always been an important part of economic and business research (Eggert et al., 2013; Wagenhofer, 2014). During the COVID pandemic, this revenue research was also focused on analysing revenue trends, and specifically, frequent revenue losses due to the pandemic's adverse effects on business and society (Chancharat and Meeprom, 2022; Kim et al., 2020; Qiang, 2022). In our literature review, we concentrate on examining existing literature sources related to high-tech technology companies and specific methods that were applied to assess their revenue trends throughout the pandemic time. For that, we have organised the review of the extant research sources into the following three streams.

The first stream includes articles that discuss the performance of high-tech and technology companies in terms of revenue and other business measurement during the COVID pandemic. Connell et al. (2022) identified a group of 32 fast-growing companies, the majority of which (15 companies, or approximately 47% of the total) were in the high-tech and technology industries, which continued to grow during the pandemic

period. Mioduchowska-Jaroszewicz et al. (2023) considered the performance of information technology companies during the COVID pandemic. Based on the study results, the authors concluded that the overall majority of these companies were able to maintain or improve their financial performance, in particular, revenue generation, during the pandemic, which would go against the general trend observed in the economy throughout the COVID pandemic.

Lozic (2021) compared the revenue and profit of ‘old’ media companies (Comcast and NY Times) and ‘new’ media companies (Netflix and Spotify) before and during the pandemic. This research demonstrated that ‘new’ media companies increased their revenue and profit during the 2020 pandemic period, while ‘old’ media companies experienced a slight reduction during the same period. Payraudeau et al. (2021) concluded that technology adoption has become a far more important performance differentiator during the COVID disruption. The authors found that in two-thirds of the industries studied, companies that embraced key disruptive technologies experienced significant revenue growth during the pandemic.

With respect to the M7 companies, we were able to find only a few literature sources that discussed some of the companies’ performance during the COVID period. According to Charles and Uford (2023), Amazon’s profitability, liquidity, solvency, and efficiency ratios were excellent throughout the pandemic. Parades-Corvalan et al. (2023) pointed out that e-commerce and the use of social media with Meta (then Twitter) grew rapidly during the COVID outbreak. Another paper (Barrantes and Leach, 2021), by analysing stock market prices, indicated that Microsoft was booming during the pandemic. Contrary to that, Lu (2023), based on Apple’s financial statements and strategy analyses, suggested that the company’s profitability did not have a dramatic improvement during the pandemic years.

Overall, the first stream of research clearly indicates that the high technology-driven companies were capable, in many cases, of maintaining or increasing their revenue throughout the beginning and subsequent years of the COVID pandemic. At the same time, we were not able to identify a literature source(s) that would consider revenue trends for all M7 companies during the COVID pandemic.

The second stream pertains to literature that discusses methods for assessing the performance of high-tech companies, particularly in relation to revenue trends during the COVID pandemic. Our literature review showed a variety of approaches, methods and tools used for this assessment. In Mioduchowska-Jaroszewicz et al. (2023), the objective of the investigation was to assess the economic status of high technology firms during the pandemic period. For this purpose, the authors utilised several analytical tools such as financial ratio analysis and descriptive statistics of revenue trends. Lozic (2021) applied a regression analysis for identifying differences in financial performance (revenue, profit, and operating margins) of certain media corporations (Comcast, NY Times, Netflix, and Spotify) before and during the pandemic. Similar regression analysis of revenue and other performance measures was used in other research papers (Hossain et al., 2023; Klöckner et al., 2023; Kramarova et al., 2022).

In these papers, the data for regression analysis combined all available historical records in pre-COVID and COVID periods. This approach might not clearly distinguish the revenue trends before the pandemic that could be used to predict the revenue trends at the pandemic disruption and the actual revenue trends throughout the same pandemic time. The difference between the predicted and actual revenue would potentially indicate loss, growth, or no change of revenue at the pandemic time. The latter approach is

referred to as counterfactual analysis, which was applied in various areas of business before the COVID outbreak (see for example, Hayashi, 2014; Baudry and Bonnet, 2019). Counterfactual analysis is a method used to evaluate causal relationships and predict outcomes by considering what would have happened in an alternative scenario, known as the counterfactual condition (Pearl, 2009). This method allows for the comparison of an actual event with a hypothetical scenario in which the event did not occur, i.e., the counterfactual scenario. It assumes that the factors influencing both the event and outcome are well understood and controlled. However, counterfactual analysis has shortcomings, including challenges in modelling complex causal relationships, difficulties in estimating unobserved variables, and reliance on assumptions that may not accurately reflect reality (Pearl, 2009; Morgan and Winship, 2015). Despite these limitations, counterfactual analysis remains an important approach for assessing revenue trends during the pandemic.

Rhee (2024) analysed the economic impact of the pandemic on the sales revenue of basic commercial areas in Seoul, Korea. The counterfactual revenue in the pandemic period was extrapolated by the Holt-Winter's model, also known as the triple exponential smoothing, including level, trend, and seasonality components in the model. This model was developed on a time series quarterly data prior to the start of the COVID pandemic in the first quarter of 2020. However, the justification for using this time series model and its accuracy level were not provided, which we consider essential elements in selecting effective forecasting models. Ehrlich et al. (2021) forecasted the 2020 revenue for each of the four major sports leagues (baseball, basketball, hockey, and football). The expected revenue during the pandemic year was identified by a regression model based on the leagues' data in 2007–2019. In this research, the validity of using the regression analysis was also not substantiated.

Another group of authors (Coelho and Duarte, 2023) utilised the time series autoregressive integrated moving average (ARIMA) model with a seasonal part for the counterfactual analysis of public debts and budget deficit in the European Union (EU). However, the justification of choosing the ARIMA model and its comparison with other alternative forecasting methods was not provided. Abe et al. (2022) employed the seasonal ARIMA (SARIMA) model to analyse the impact of COVID pandemic on fishing industry in Japan. The difference between the SARIMA model's prediction and actual results was used to identify the revenue trends throughout the pandemic time.

Overall, the review of papers in the second stream demonstrated that counterfactual analysis was an important instrument to assess revenue trends (loss, increase, or no change) during the COVID pandemic. However, the authors using this analysis typically applied a single time series model to estimate the revenue during the pandemic period without its proper validation. In addition, we were not able to identify a research paper that would apply counterfactual analysis towards estimating revenue trends for the M7 companies.

The third and final stream pertains to the improvement of time series forecasting in counterfactual analysis during the COVID pandemic by utilising ensemble methods. This approach, by combining several forecasting models together, leads to an increase in their robustness in prediction and an overall improvement in forecast accuracy (Dutta and Marodia, 2015; Hyndman and Athanasopoulos, 2021; Shmueli et al., 2020; Zhou, 2012).

The ensemble methods were also used in some COVID related research. By utilising the ensemble model approach, Ospina et al. (2023) effectively reduced forecasting errors

and thus demonstrated that this approach is a promising methodology for estimating COVID effects, improving data forecasting accuracy, and providing rapid responses in the early stages of the pandemic. Babski-Reeves et al. (2023) recognised that an ensemble model that combines four individual time series models was effectively used to predict, with a high accuracy level, the number of COVID cases in the state of California.

Rodrigues et al. (2021) utilised different forecasting methods as well as the ensemble model that combines all these methods to predict consumption of fuels for vehicles in Brazil before and during the COVID pandemic. The research results indicated that the ensemble model improved forecast accuracy compared to individual forecasting methods. Lee and Sul (2021) and Roberts (2021) reported similar advantages of ensemble methods in estimating the effects of COVID during the pandemic. Overall, the literature review in the third stream reveals that the ensemble methods in time series forecasting can improve the accuracy of revenue estimation during the COVID pandemic. However, we were not able to locate any research paper that would apply an ensemble method(s) in analysing revenue trends of the M7 companies throughout the pandemic time.

In summary, we have drawn several important conclusions from the literature review. First, this review showed that the high-tech companies were able to maintain or increase their revenue throughout the COVID pandemic. Second, the existing literature sources established that counterfactual analysis was an important instrument to assess revenue trends (loss, increase, or no change) during the pandemic time. Third, the literature review recognised that the utilisation of time series ensemble models to estimate revenue trends through the pandemic is more robust and accurate than the usage of individual time series methods. Finally, the authors of this paper could not identify a single research paper that would estimate and analyse revenue trends of the M7 companies throughout the COVID pandemic.

3 Methodology

Based on the results of the literature review, we formulated the following three research objectives:

- 1 Develop the most accurate time series forecasts for the Magnificent Seven (M7) companies' revenues throughout the COVID pandemic.
- 2 Construct an ensemble method forecast for each M7 company's revenue and apply counterfactual analysis to obtain and evaluate estimated revenue trends (loss, increase, or no change) during the COVID pandemic.
- 3 Identify whether revenue trends in the M7 companies were adversely affected by the COVID pandemic.

To address the designated research objectives, we collected quarterly revenue data from 2010 to 2023 for all M7 companies (Macrotrends.net, 2024) with the exception of Tesla. For the latter company, the collected dataset contained quarterly revenues from 2014 to 2023 (Macrotrends.net, 2024). This was due to the fact that Tesla's Model S, the first mass-produced vehicle in the company, went into full production and delivery to customers in June of 2012, and it did not reach a reasonably high production level until the end of 2013 (Reed and Diongson, 2024; Wu, 2024).

To consider research objective 1, we utilised the historical data for the pre-COVID period, i.e., quarterly data for the M7 companies from 2010 to 2019 (2014 to 2019 for Tesla) to develop time series forecasting models for each M7 company. The models' selection depended on the preliminary analysis of the historical data and identified, out of this analysis, respective data patterns, e.g., trend and/or seasonality, for each dataset. This will be discussed in the next section of this paper.

For each M7 company, we identified the three best time series models based on the two measures of forecast accuracy: root mean squared error (RMSE) and mean absolute percentage error (MAPE). Both of them are most commonly used accuracy measures in analysing the performance of time series model (Hyndman and Athanasopoulos, 2021). Although either measure can be used separately without applying the other measure, by utilising them together, these measures can complement each in identifying the most accurate forecasting methods. Therefore, in this research we applied them together to recognise the three best forecasting models and to combine them into ensembles.

To address research objective 2, we employed the three best forecasting models to identify an ensemble time series model for each M7 company. For that, we calculated the respective weight for each of the three best time series in the combined ensemble model. A weight for model i and M7 company j (w_{eij}) was identified using the following formulae:

$$w_{1ij} = \frac{\frac{1}{RMSE_{ij}}}{\frac{1}{RMSE_{1j}} + \frac{1}{RMSE_{2j}} + \dots + \frac{1}{RMSE_{ij}}}, i = 1, 2, 3; j = 1, 2, \dots, 7 \quad (1)$$

$$w_{2ij} = \frac{\frac{1}{MAPE_{ij}}}{\frac{1}{MAPE_{1j}} + \frac{1}{MAPE_{2j}} + \dots + \frac{1}{MAPE_{ij}}}, i = 1, 2, 3; j = 1, 2, \dots, 7 \quad (2)$$

$$w_{eij} = \frac{1}{2}(w_{1ij} + w_{2ij}) \quad (3)$$

where

- w_{1ij} , w_{2ij} – represent weights for the i^{th} best time series model based on RMSE and MAPE, respectively, and for M7 company j
- $\frac{1}{RMSE_{ij}}$ and $\frac{1}{MAPE_{ij}}$ – denote reciprocal (multiplicative inverse) values for i^{th} best times series model's RMSE and MAPE, respectively, and for M7 company j .

With the constraint that the total weight is equal to 1, $\sum_{i=1}^3 w_{eij} = 1$, for the ensemble model's weights. The reason for using reciprocal values of RMSE and MAPE to identify weights is based on the nature of these measures. The lower the value of them, the better the model's forecast accuracy, and therefore, the higher the weight for this forecasting model should be. To correctly reflect the opposite relationships between the model's

weights and its respective RMSE and MAPE measures, the latter measures are converted in the reciprocal format.

The ensemble forecast F_{ejt} for M7 company j and quarter t was identified as the weighted mean of the three best forecasting methods:

$$F_{ejt} = \sum_{i=1}^3 w_{eij} F_{ijt}, i = 1, 2, 3 \text{ and } j = 1, 2, \dots, 7 \quad (4)$$

where

- w_{eij} – represents an ensemble weight for the i^{th} best time series model for the j^{th} M7 company
- F_{ijt} – denotes revenue forecast for the i^{th} best time series model for the j^{th} M7 company in period t .

Based on the developed ensemble models, we identified the quarterly forecasts for each M7 company during COVID periods. We then utilised counterfactual analysis to compare the actual revenue with the respective forecast in each quarters t during the period of 2020–2023 and for each M7 company j ($j = 1, 2, \dots, 7$), and also to calculate their difference (revenue change) R_{cjt} and proportion (percentage) of change P_{cjt} in the respective revenue forecast F_{ejt} :

$$R_{cjt} = A_{jt} - F_{ejt} \quad (5)$$

$$P_{cjt} = \frac{R_{cjt}}{F_{ejt}} \quad (6)$$

where

- F_{ejt} – denotes an ensemble model's revenue forecast in equation (4) for the j^{th} M7 company in quarter t during the period of 2020–2023
- A_{jt} – represents an actual revenue for the j^{th} M7 company in quarter t during the period of 2020–2023.

Using formulae (5)–(6), we also calculated annual, two-year, and four-year forecasts for each M7 company from 2020 to 2023. We also explained the results of counterfactual analysis with the tables and plots that contain revenue trends. These revenue trends could be negative (indicating a loss), positive (indicating an increase), or show no significant trend during the COVID period. The analysis described led to research objective 3: to assess whether revenue trends in the M7 companies were adversely affected by the pandemic in the years 2020–2023.

To apply time series models in this research, we utilised various R software packages, particularly the forecast package. The latter enabled us to utilise time series methods described in this section, identify the ensemble models, and perform counterfactual analysis of revenue trends during the COVID time periods.

4 Analysis, results, and discussion

4.1 Preliminary analysis

To address all research objectives, we began by analysing the quarterly revenue data for each M7 company from 2010 to 2023 (2014 to 2023 for Tesla). This analysis revealed that the quarterly data contained no missing values or unusual outliers, making it suitable for time series analysis and forecasting. After this step was completed, we continued the preliminary analysis by identifying autocorrelation coefficients for the first four lags of the quarterly M7 data (Figure 1).

Figure 1 Autocorrelation charts for M7 companies' quarterly revenues from 2010 to 2023

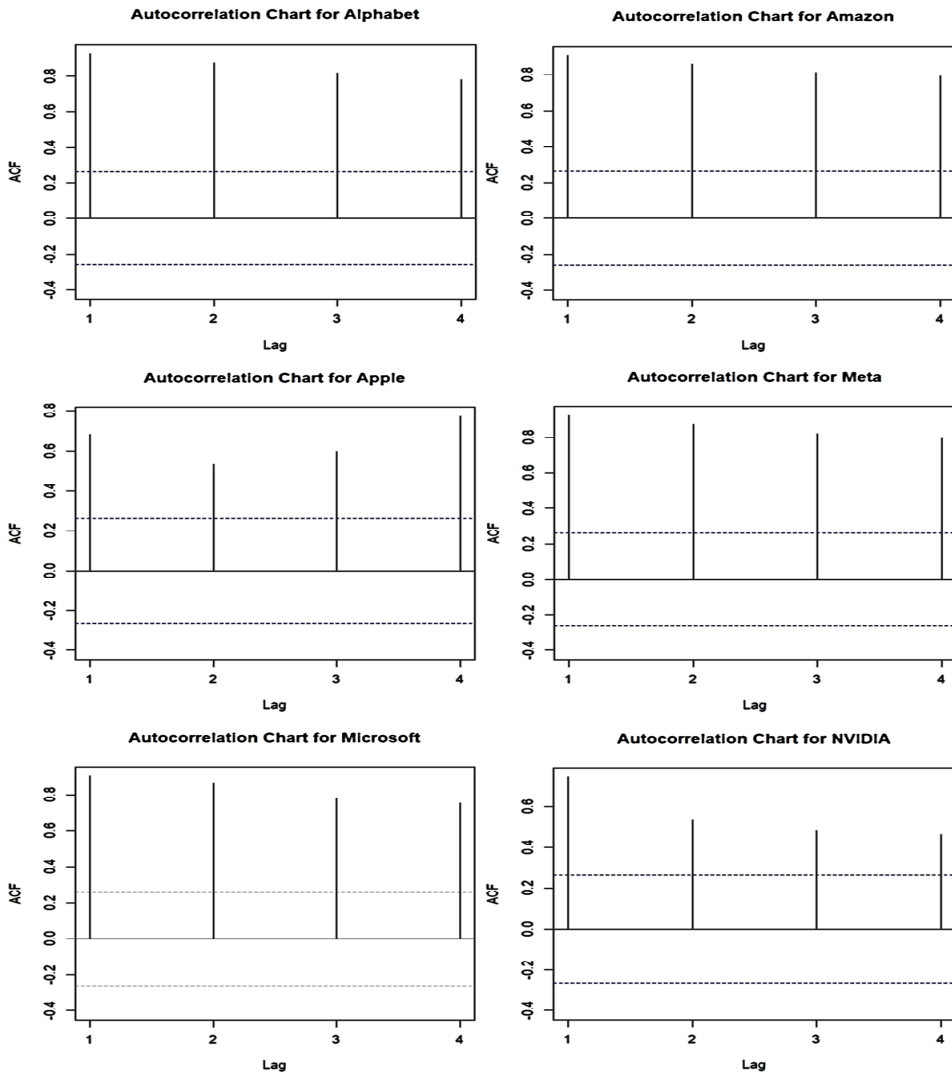
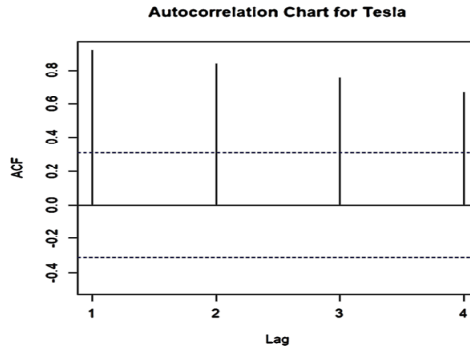


Figure 1 Autocorrelation charts for M7 companies' quarterly revenues from 2010 to 2023 (continued)



As can be seen from the charts in Figure 1, all autocorrelation coefficients were positive and statistically significant, i.e., above the dotted horizontal line in the charts representing the margin of statistical significance. Statistically significant and strong autocorrelation coefficients in lag 1 were indicative of a trend pattern in all M7 date sets. At the same time, statistically significant and strong autocorrelation coefficients in lag 4 revealed quarterly seasonality in all M7 datasets. Therefore, these results demonstrated that each M7 company's revenue data included trend and seasonality patterns.

4.2 Best time series models

According to objective 1 of this research, we continued our analysis by identifying the best forecasting method for each M7 company. As discussed in the Methodology section, for building time series forecasting models for all M7 companies (except for Tesla), we utilised the quarterly data for the pre-COVID period from 2010 to 2019. For Tesla, the pre-COVID period included the quarterly data from 2014 to 2019. Assuming the previously identified trend and seasonality data patterns for all M7 companies, we selected the following four groups of times series methods that are commonly applied for the historical datasets with trend and seasonality:

- Regression models with trend and seasonality.
- Two-level models that combine a regression model with trend and seasonality (level 1) and a trailing moving average (MA) for the regression model residuals (level 2).
- Holt-Winter's (HW) exponential smoothing models (triple exponential smoothing), which incorporate error, trend, and seasonality patterns.
- ARIMA models with trend and seasonal options.

The specified groups of models are considered quite effective in developing forecasts for data with trend and seasonality and are well discussed in literature and practitioner sources (Montgomery et al., 2016; Shmueli et al., 2020). We identified several alternative models in each recognised group of models, ran these models, and then compared their accuracy using RMSE and MAPE measures. Based on this comparison, we selected the three best time series models for each M7 company by utilising the lowest values of RMSE and MAPE criteria of forecast accuracy (Table 1).

Model 1 in the table represents a two-level model with quadratic trend and seasonality regression (level 1) and MA for the regression residuals. Model 2 denotes an HW exponential smoothing model, and model 3 signifies an ARIMA model with trend and seasonality options. The fourth group of time series methods applied, i.e., regression model with trend and seasonality, have produced a significantly higher RMSE and MAPE for each M7 dataset, and thus, was not considered further in our analysis. Overall, based on the values of MAPE in Table 1, we have achieved a significant accuracy between 1.85% and 10.11% for six of the M7 companies, and also a relatively good forecasting accuracy of 10.52 to 12.09% for Tesla. The latter may be due to utilising a smaller number of historical data points for Tesla, i.e., 24 quarterly revenues for the period of 2014–2019, versus all other M7 companies with a larger number of 40 quarterly revenues from 2010 to 2019.

Table 1 RMSE and MAPE accuracy measures for three best models

<i>M7 company</i>	<i>RMSE for best models, \$Mln</i>			<i>MAPE for best models, %</i>		
	<i>Model 1</i>	<i>Model 2</i>	<i>Model 3</i>	<i>Model 1</i>	<i>Model 2</i>	<i>Model 3</i>
Alphabet	698.31	939.39	831.16	2.78	3.34	3.12
Amazon	1,245.37	1,593.38	962.40	3.95	3.26	1.85
Apple	3,307.77	4,467.32	2,894.91	6.47	6.75	3.75
Meta	312.01	166.86	169.18	10.11	3.11	1.85
Microsoft	652.98	1,279.34	1,253.18	2.13	4.66	3.90
NVIDIA	107.54	215.36	153.52	4.40	7.81	4.82
Tesla	407.71	891.60	814.84	10.52	12.09	11.52

While developing the three best time series models, the parameter configurations of those models, specifically for HW (model 2) and ARIMA (model 3) options, were somewhat different. The various configurations of the best time series models for each M7 company are presented in Appendix to this paper.

4.3 Ensemble forecasting models and counterfactual analysis

To fulfil research objective 2, we developed, as discussed in the methodology section, the ensemble model for each M7 company based on the respective best three models that were described in the previous subsection. To do that, we first developed weights for these models using the formulae (1)–(3). These weights for each M7 company are presented in Table 2.

As calculated by formulae (1)–(3), each weight in Table 2 is reflective of the respective model's values of RMSE and MAPE but in a reciprocal way, i.e., the lower the values of RMSE and MAPE the higher the weight for a respective time series model. Contrary to that, the higher the values of RMSE and MAPE for a respective model, the lower the associated weight. This way we would ensure that more accurate models will have a higher weighted representation in the ensemble combination. As can be also seen from Table 2, the total weights for each M7 ensemble model is equal to 1.

Ensemble forecasts for the M7 companies are identified using formula (4) as a weighted sum of the best models' forecasts in the respective periods. Based on the quarterly revenue data from 2010 to 2019 (2014 to 2019 for Tesla), we produced

quarterly ensemble forecasts for each M7 company and identified their respective accuracy measures presented in Table 3.

Table 2 Ensemble weights for best time series models

<i>M7 company</i>	<i>Weights for ensemble models</i>			<i>Total weight</i>
	<i>Model 1</i>	<i>Model 2</i>	<i>Model 3</i>	
Alphabet	0.3773	0.2965	0.3262	1.0000
Amazon	0.2775	0.2663	0.4562	1.0000
Apple	0.3091	0.2587	0.4322	1.0000
Meta	0.1575	0.3656	0.4769	1.0000
Microsoft	0.4956	0.2397	0.2647	1.0000
NVIDIA	0.4292	0.2271	0.3437	1.0000
Tesla	0.4350	0.2731	0.2919	1.0000

The results in Table 3 show that the ensemble methods produced high accuracy performance measures, with the MAPE varying between 1.79% for Microsoft (the best MAPE) to 9.96% for Tesla. Comparing the values in Table 3 with the similar values in Table 1, we can conclude that the RMSE and MAPE accuracy measures for ensemble models in Table 3 are either very close or even sometime better (for example, for Amazon, Tesla, and Meta) than the measures for the respective individual time series models. This also confirms the conclusion that was made in the literature review section that ensemble models could be more robust and accurate than the usage of individual time series methods.

Table 3 RMSE and MAPE accuracy measure for M7 ensemble models

<i>M7 company</i>	<i>Accuracy measures for ensemble models</i>	
	<i>RMSE, \$Mln</i>	<i>MAPE, %</i>
Alphabet	711.63	2.81
Amazon	866.17	1.79
Apple	2,638.73	4.32
Meta	140.26	2.48
Microsoft	924.29	3.17
NVIDIA	139.73	5.04
Tesla	641.40	9.96

We utilised the developed ensemble models for the counterfactual analysis of M7 companies' revenues during the COVID periods. For that, we applied formula (5) to identify revenue changes (differences) between ensemble forecasts and actual revenues for each M7 company in quarterly periods of 2020–2023 (Figure 2). The plots in Figure 2 clearly showed that for quarterly periods starting from the end of 2020 and to the end of 2023, the actual revenue consistently exceeded the estimated revenue with the positive revenue changes (above the horizontal dotted line with zero change). Contrary to that, in some M7 companies like Alphabet, Apple, Meta, and Tesla, the revenue changes in the first several quarters of 2020 were negative (below the horizontal dotted line of zero

change), which indicated that the actual revenue was lower than the estimated revenue based on ensemble forecast.

Figure 2 Actual and forecasted revenue, and revenue changes for M7 companies during COVID pandemic (quarter periods from 2020 to 2023)

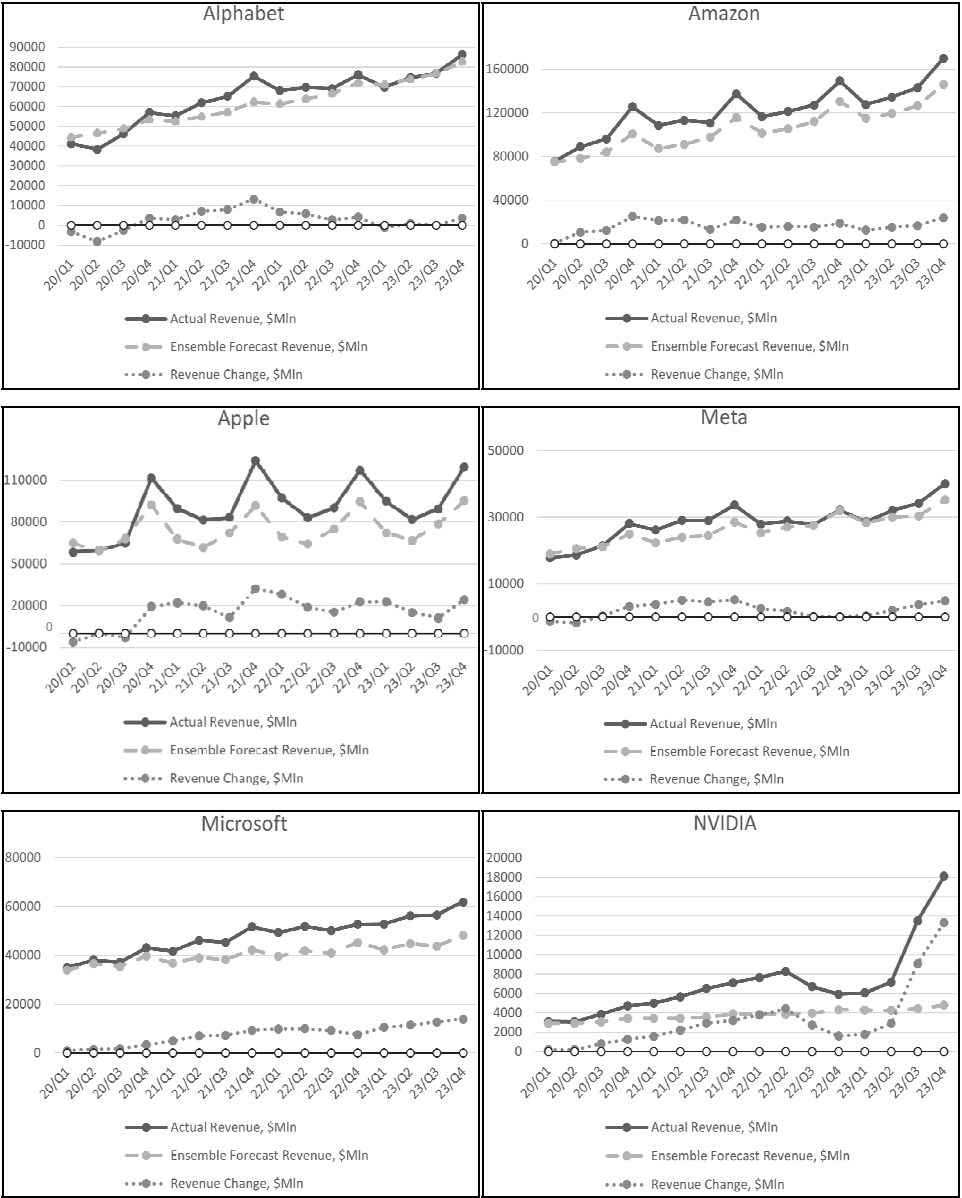
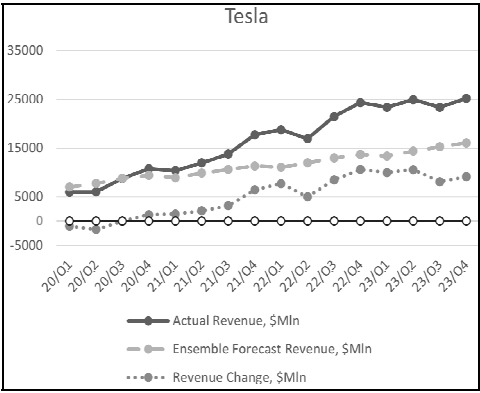


Figure 2 Actual and forecasted revenue, and revenue changes for M7 companies during COVID pandemic (quarter periods from 2020 to 2023) (continued)



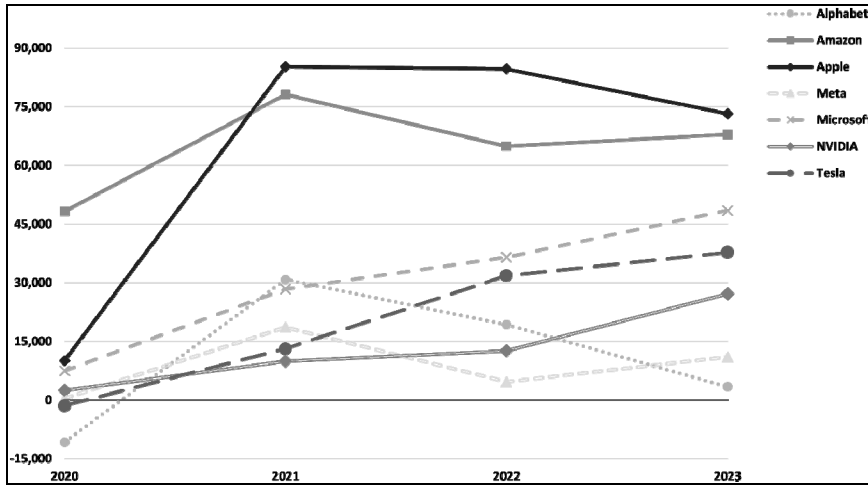
4.4 Revenue trends (changes) during COVID pandemic

The results of the counterfactual analysis were summarised by annual revenue changes in each year during the period of 2020–2023, by two-year periods of 2020–2021 and 2022–2023, and for the entire four years of COVID (Table 4). Plots of annual revenue changes for the M7 companies can be seen in Figure 3.

Table 4 Annual, two-year, and four-year revenue trends in M7 companies from 2020 to 2023

Years	Revenue changes in M7 companies, \$Mln						
	Alphabet	Amazon	Apple	Meta	Microsoft	NVIDIA	Tesla
2020	−10,651	48,294	10,125	391	7,506	2,450	−1,423
2021	30,855	78,114	85,191	18,601	28,353	9,941	13,149
2022	19,288	64,920	84,687	4,669	36,483	12,551	31,746
2023	3,418	67,911	73,194	10,976	48,491	27,124	37,714
Total in 2020–2021	20,203	126,409	95,316	18,992	35,860	12,390	11,726
Total in 2022–2023	22,706	132,831	157,881	15,646	84,974	39,675	69,460
Total in 2020–2023	42,910	259,239	253,197	34,638	120,834	52,065	81,186

The data in Table 4 and associated plots in Figure 3 showed that for five of the M7 companies (Amazon, Apple, Meta, Microsoft, and NVIDIA), the annual revenue changes from 2020 to 2023 were consistently positive. This means that, according to counterfactual analysis, the actual revenue continued to exceed the revenues estimated by the ensemble forecasting models. The two of the M7 companies, Alphabet and Tesla, showed a negative revenue change in 2020, and thus sustained revenue loss according to counterfactual analysis. However, the combined revenue trend (changes) during the period of 2020–2021 were positive for all M7 companies, including Alphabet and Tesla, and continued to be positive in the subsequent years of 2022–2023, and all for the whole four-year period.

Figure 3 Annual revenue changes in M7 companies from 2020 to 2023, \$Mln

To better understand the dynamic of revenue trends, we also identified, using formula (6), percentages of estimated annual revenue changes during COVID, as well as for the two-year periods of 2020–2021 and 2022–2023, and for the entire four-year period of 2020–2023. These results are summarised in Table 5 and Figure 4.

Table 5 Annual, two-year, and four-year revenue changes in M7 companies from 2020 to 2023, %

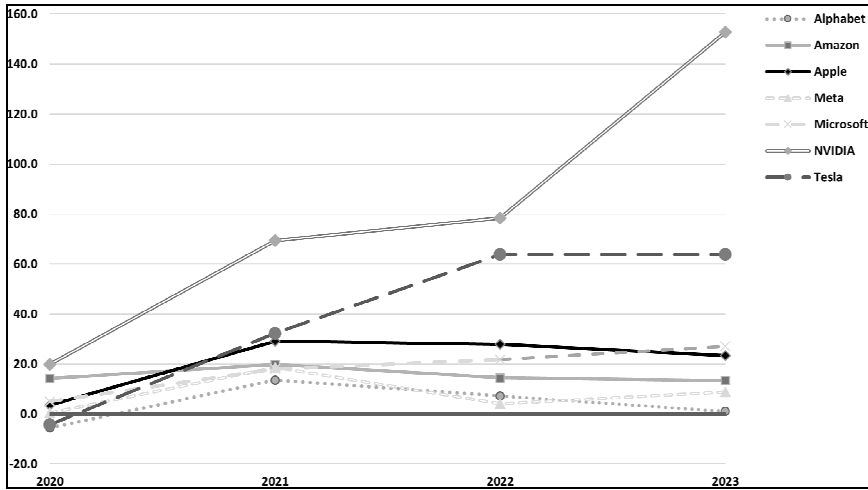
Years	Revenue trend in M7 companies in 2020–2023, %						
	Alphabet	Amazon	Apple	Meta	Microsoft	NVIDIA	Tesla
2020	–5.5	14.3	3.6	0.5	5.1	19.9	–4.3
2021	13.6	19.9	29.1	18.7	18.1	69.4	32.3
2022	7.3	14.5	28.0	4.2	21.8	78.4	63.9
2023	1.1	13.4	23.4	8.9	27.1	152.8	63.9
Total in 2020–2021	4.8	17.3	16.5	10.3	11.9	46.5	15.9
Total in 2022–2023	4.0	13.9	25.7	6.6	24.5	117.5	63.9
Total in 2020–2023	4.3	15.4	21.2	8.2	18.6	86.2	44.5

As can be seen from Table 5 and Figure 4, most of the companies (except for Alphabet and Tesla in 2020) showed a consistently positive percentage of revenue changes during the pandemic years of 2020 and 2021, and subsequent years of 2022–2023. One interesting and unusual result from the data in Table 5 and Figure 4 was that four M7 companies including Alphabet, Amazon, Apple, and Meta, have had the highest positive percentage of revenue growth in 2021, the second year of the COVID pandemic. The other three M7 companies – Microsoft, NVIDIA, and Tesla – consistently increased these percentages throughout the COVID time periods of 2020–2023. Overall, the ultimate conclusion is that the COVID pandemic led to an increase in the revenue of the M7 companies from 2020 to 2023.

Based on the discussed results of revenue trends in the M7 companies, we can now assess the validity of objective 3 of this research concerning the revenue trend of the M7

companies throughout the COVID pandemic. Most of the M7 companies (Amazon, Apple, Meta, Microsoft, and NVIDIA) were able to sustain and increase their revenue versus estimated ensemble forecasts during each year of the COVID pandemic in 2020–2023. The other two M7 companies (Alphabet and Tesla), although losing revenue in 2020, which is most likely due to COVID-related effects, were still able to increase their revenue in the two-year period of 2020–2021, the two main years of the COVID pandemic, and in the subsequent years of 2022–2023.

Figure 4 Annual revenue trend in M7 companies from 2020 to 2023, %



Summarising these results, we can conclude that for a short-term period, e.g., within the first pandemic year of 2020, some of the M7 companies (2 out of 7) have experienced revenue losses and thus were adversely affected by the COVID pandemic. Overall, however, for the two-year COVID period of 2020–2021 and subsequent years of 2022–2023, all M7 companies' revenue trends, i.e., continued revenue growth, were not adversely affected by the COVID pandemic.

5 Conclusions

5.1 Goals and methodology

Our main goals of this research were to identify and analyse revenue trends of the M7 companies during the COVID pandemic and to evaluate whether these trends were adversely affected by the pandemic.

Through literature review, we revealed that high-tech companies' revenue was mainly not adversely affected by the COVID pandemic and continued to grow in these companies throughout the pandemic periods. However, we could not identify a single research paper or publication that would estimate and analyse revenue trends of the M7 companies throughout the COVID pandemic. By examining existing literature sources, we also established that counterfactual analysis was an important instrument to assess the revenue trends (loss, increase, or no change) during the pandemic periods. In addition,

our literature review recognised that the utilisation of time series ensemble models to forecast revenue trends throughout the pandemic would be more robust and accurate than the usage of individual time series forecasting methods.

Our research methodology was based on the three objectives stated in the methodology section of this paper. They include:

- 1 develop the most accurate time series forecasts for the Magnificent Seven (M7) companies' revenues throughout the COVID pandemic
- 2 construct an ensemble method forecast for each M7 company's revenue and apply counterfactual analysis to obtain and evaluate estimated revenue trends (loss, increase, or no change) throughout the COVID pandemic
- 3 identify whether the revenue trends in the M7 companies were adversely affected by the COVID pandemic.

The research datasets contained quarterly revenue data from 2010 to 2023 for each M7 company, except for Tesla with quarterly revenue from 2014 to 2023. To fulfil the research objectives, we developed a variety of time series forecasting methods for the data with trend and seasonality including regression, two-level forecasting, Holt-Winter's and ARIMA models. Based on the three most accurate time series models for each M7 company, we created an ensemble forecasting model that was employed in counterfactual analysis to identify revenue trends (changes) for the M7 companies at the COVID time and recognise if these revenue trends were adversely affected by the COVID pandemic.

5.2 Research contributions

By utilising the research methodology with the specified methods and R software, we achieved a number of important results in this research that contribute to the theory and practice of analysing revenue trends of companies during the COVID onslaught. First, we conducted a novel study which explored the effect of the COVID pandemic on the revenue trends of the M7 companies. Second, for each M7 company, we creatively utilised the time series ensemble model to estimate quarterly revenue values during the COVID time periods. This ensemble model employed the three most accurate forecasting models identified for each M7 company with the pre-COVID datasets, and the ensemble weights for these models calculated by the reciprocal values of their RMSE and MAPE accuracy measures. Third, we innovatively applied these ensemble forecasting models for counterfactual analysis of revenue trends for each M7 company in the quarter periods during the COVID pandemic. We also analysed revenue trends on the annual, two-year and four-year periods from 2020 to 2023 by utilising the estimated values and percentages of revenue changes.

Fourth, we determined that the COVID pandemic led to an increase in the revenue of the M7 companies from 2020 to 2023. Five of the M7 companies (Amazon, Apple, Meta, Microsoft, and NVIDIA) sustained and increased their revenues compared to ensemble models' forecasts during each year of the pandemic. The other two M7 companies (Alphabet and Tesla), despite experiencing revenue losses in 2020, were able to surpass their forecasted revenues in the two-year periods of 2020–2021 and 2022–2023. Fifth, we concluded that, for a short-term period – specifically in the first year of the COVID pandemic in 2020 – some of the M7 companies may have experienced revenue losses due

to the initial adverse effects on demand for their products. However, over the longer two-year period from 2020 to 2021 and the subsequent years of 2022 to 2023, revenue trends for all M7 companies were not adversely affected by the pandemic. Finally, this research provided specific directions on how to develop a new methodology and practical application of analysing revenue trends for any organisation during the COVID pandemic.

5.3 Limitations and future research directions

Despite the extensive analysis, numerous contributions, and practical results of this research, it still has some limitations. We focused our study to analysing revenue trends, which is an important performance measure directly associated with demand, profit, stock price, and other metrics for all M7 companies. However, we did not examine additional performance metrics, such as profit and stock price, throughout the pandemic.

In this research, we applied counterfactual analysis, which has several limitations, including challenges in modelling complex causal relationships, difficulties in estimating unobserved variables, and reliance on assumptions that may not accurately reflect reality. We also utilised the most commonly used time series analysis methods without exploring other machine learning techniques, such as neural networks and bootstrap ensembles.

The limitations described could serve as a foundation for future research expansion. For example, we could analyse not only revenue trends but also the profitability and stock market prices of the M7 companies during the COVID pandemic. Additionally, we could apply other machine learning techniques, such as neural networks and bootstrap ensembles, alongside the ensemble methods utilised in this study.

Declarations

All authors declare that they have no conflicts of interest.

References

- Abe, K., Ishimura, G., Baba, S., Yasul, S. and Nakamura, K. (2022) 'Evaluating the impact of COVID-19 on ex-vessel prices using time-series analysis', *Fisheries Science*, Vol. 88, pp.191–202, <https://doi.org/10.1007/s12562-021-01574-x>.
- Babski-Reeves, K., Eksioglu and Hampton, D. (2023) 'Analysis of machine learning and ensemble methods for forecasting COVID-19 cases in California', *Proceedings of the IISE Annual Conference & Expo 2023*, pp.1–7, https://doi.org/10.21872/2023IISE_1464.
- Barrantes, R. and Leach, T. (2021) 'The impact of the COVID-19 pandemic on the one of the biggest technology giant shares: a financial analysis on Microsoft Corporation', *Journal of Student Research*, Vol. 10, No. 3, pp.1–16, <https://doi.org/10.47611/jsrhs.v10i3.1689>.
- Baudry, M. and Bonnet, C. (2019) 'Demand-pull instruments and the development of wind power in Europe: a counterfactual analysis', *Environmental and Resource Economics*, Vol. 73, pp.385–429, <https://doi.org/10.1007/s10640-018-0267-3>.
- Chancharat, S. and Meeprom, S. (2022) 'The effect of the COVID-19 outbreak on hospitality and tourism stock returns in Thailand', *Anatolia*, Vol. 33, No. 4, pp.564–575, <https://doi.org/10.1080/13032917.2021.1982738>.
- Charles, I.I. and Uford, I.C. (2023) 'Comparative analysis and evaluation of business and financial performance of Amazon.com: a three-year period critical review of exceptional success', *European Journal of Business, Economics and Accountancy*, Vol. 11, No. 2, pp.69–92.

- Coelho, J.A. and Duarte, A.P. (2023) 'The European fiscal framework: counterfactual analysis to its compliance in the hypothetical scenario without the COVID-19 pandemic', *European Journal of Comparative Economics*, Vol. 20, No. 2, pp.265–298, <http://dx.doi.org/10.25428/1824-2979/025>.
- Connell, C.M., Lemyze, C. and McGill, W.L. (2022) 'Firms that prosper in all weathers: surviving recessions and plagues', *Journal of Business Strategy*, Vol. 43, No. 2, pp.135–147, <https://dx.doi.org/10.1108/JBS-10-2020-0247>.
- Dutta, G. and Marodia, D.P. (2015) 'Comparison of forecasting techniques in revenue management for a national railway in an emerging Asian economy', *International Journal of Revenue Management*, Vol. 8, No. 2, pp.130–152, <https://doi.org/10.1504/IJRM.2015.070000>.
- Eggert, A., Hogreve, J., Ulaga, W. and Muenkhoff, E. (2013) 'Revenue and profit implications of industrial service strategies', *Journal of Service Research*, Vol. 17, No. 1, pp.23–39, <https://doi.org/10.1177/1094670513485823>.
- Ehrlich, J.A., Ghimire, S., Khraiche, M. and Raza, M.F. (2021) 'COVID-19 countermeasures, sporting events, and the financial impacts to the North American leagues', *Managerial Finance*, Vol. 46, No. 7, pp.887–895, <https://doi.org/10.1108/MF-05-2020-0275>.
- Hayashi, T. (2014) 'Is it abenomics or post-disaster recovery? A counterfactual analysis', *International Advances in Economic Research*, Vol. 20, pp.23–31, <https://dx.doi.org/10.1007/s11294-013-9450-z>.
- Hlavka, J. and Rose, A. (2023) *COVID-19's Total Cost to the U.S. Economy Will Reach \$14 Trillion by End of 2023* [online] <https://healthpolicy.usc.edu/article/covid-19s-total-cost-to-the-economy-in-us-will-reach-14-trillion-by-end-of-2023-new-research/> (assessed 12 June 2024).
- Hossain, A.T., Masum, A-A. and Xu, J. (2023) 'COVID-19, a blessing in disguise for the tech sector: evidence from stock price crash risk', *Research in International Business and Finance*, Vol. 65, No. 101938, pp.1–15, <https://doi.org/10.1016/j.ribaf.2023.101938>.
- Hyndman, R.J. and Athanasopoulos, G. (2021) *Forecasting: Principles and Practice*, 3rd ed., OTexts, Melbourne, Australia [online] <https://otexts.com/fpp3> (assessed 24 June 2024).
- Kim, O.S., Parker, J.A. and Schoar, A. (2020) *Revenue Collapses and the Consumption of Small Business Owners in the Early Stages of the COVID-19 Pandemic*, National Bureau of Economic Research, Working Paper, No. 28151, pp.1–56, <http://www.nber.org/papers/w28151>.
- Klößner, M., Schmidt, C.G. and Wagner, S.M. (2023) 'The COVID-19 pandemic and shareholder value: impact and mitigation', *International Journal of Production Research*, Vol. 61, No. 8, pp.2470–2492, <https://doi.org/10.1080/00207543.2022.2147235>.
- Kramarova, K., Svabova, L. and Gabrikova, B. (2022) 'Impacts of the COVID-19 crisis on unemployment in Slovakia: a statistically created counterfactual approach using the time series analysis', *Equilibrium, Quarterly Journal of Economics and Economic Policy*, Vol. 17, No. 2, pp.343–389, <https://doi.org/10.24136/eq.2022.012>.
- Lee, Y. and Sul, D. (2021) *Depth-Weighted Forecast Combination: Application to COVID-19 Cases*, Center for Policy Research Working Papers, No. 238, pp.1–10 [online] <https://ideas.repec.org/p/max/cprwps/238.html> (assessed 15 June 2024).
- Lozic, J. (2021) 'Comparison of financial indicators and the trend of the number of users of the old and new media industry during the COVID-19 pandemic', *70th International Scientific Conference on Economic and Social Development*, 25–26 June, pp.1150–1159.
- Lu, Y. (2023) 'Financial statement analysis and revenue forecast for Apple Inc.', *SHS Web of Conferences: ICDEBA 2023*, pp.1–8, <https://doi.org/10.1051/shsconf/202418102020>.
- Macrotrends.net (2024) *Macrotrends – The Premier Research Platform for Long Term Investors* [online] <https://macrotrends.net> (assessed April–May 2024).
- Mioduchowska-Jaroszewicz, E., Chybowska, W., Jedynak, B. and Magac, M. (2023) 'Accrual and cash financial performance of information technology companies during the COVID-19 pandemic', *Folia Oeconomica Stetinensia*, Vol. 23, No. 2, pp.275–293.

- Mitra, M. (2024) *What Are the Magnificent Seven Stocks?*, WSJ [online] <https://www.wsj.com/buyside/personal-finance/magnificent-7-stocks-0d68cab6> (assessed 13 June 2024).
- Montgomery, D.C., Jennings, C.L. and Kulachi, M. (2016) *Introduction to Time Series Analysis and Forecasting*, 2nd ed., John Wiley & Sons, Inc., Hoboken, New Jersey.
- Morgan, S.L. and Winship, C. (2015) *Counterfactuals and Causal Inference: Methods and Principles for Social Research*, 2nd ed., Cambridge University Press, New York.
- Nair, G.K. (2021) 'Data driven pricing strategies for hotels during the COVID-19 pandemic', *International Journal of Revenue Management*, Vol. 12, Nos. 1/2, pp.34–55, DOI: 10.1504/IJRM.2021.114966.
- Ospina, R., Oliveira, J., Ferraz, C., Leite, A. and Gondim, J. (2023) 'Ensemble algorithms to improve COVID-19 growth curve estimates', *Stats*, Vol. 6, pp.990–1007, <https://doi.org/10.3390/stats6040062>.
- Palazzo, B. (2023) *Corporate Profits in the Aftermath of COVID-19*, FEDS Notes [online] <https://www.federalreserve.gov/econres/notes/feds-notes/corporate-profits-in-the-aftermath-of-covid-19-20230908.html> (assessed 15 June 2024).
- Parades-Corvalan, D., Pezoa-Fuentes, C., Silva-Rojas, G., Rojas, I.V. and Castillo-Vergara, M. (2023) 'Engagement of the e-commerce industry in the US, according to Twitter in the period of the COVID-19 pandemic', *Heliyon*, Vol. 9, pp.1–30, <https://doi.org/10.1016/j.heliyon.2023.e16881>.
- Payraudeau, J-S., Marshall, A., Jacob, D. and Handa, R. (2021) 'How tech-savvy organizations are outperforming their peers during the pandemic', *Strategy and Leadership*, Vol. 49, No. 2, pp.29–35, <https://dx.doi.org/10.1108/SL-02-2021-0018>.
- Pearl, J. (2009) *Causality: Models, Reasoning, and Inference*, 2nd ed., Cambridge University Press, New York, USA.
- Qiang, M. (2022) 'Quantifying the loss of China's tourism revenue induced by COVID-19', *Current Issues in Tourism*, Vol. 25, No. 24, pp.3919–3924, <https://doi.org/10.1080/13683500.2022.2067524>.
- Reed, E. and Diongson, D. (2024) *History of Tesla and its Stock: Timeline, Facts & Milestones*, TheStreet [online] <https://www.thestreet.com/technology/history-of-tesla-15088992> (assessed 10 July 2024).
- Rhee, H. (2024) 'Assessing the economic impact of COVID-19 through a counterfactual analysis', *East Asian Economic Review*, Vol. 28, No. 1, pp.69–94, <https://dx.doi.org/10.11644>.
- Roberts, S. (2021) *All Together Now: The Most Trustworthy COVID-19 Model is an Ensemble*, Technology Review [online] <http://proxylib.csueastbay.edu/login?url=https://www.proquest.com/other-sources/alltogether-now-most-trustworthy-covid-19-model/docview/2820956998/se-2?accountid=28458> (assessed 15 June 2024).
- Rodrigues, L., Rodrigues, L. and Bacchi, M.R.P. (2022) 'Performance evaluation of forecasting models based on time series and machine learning techniques: an application to light fuel consumption in Brazil', *International Journal of Energy Sector Management*, Vol. 16, No. 4, pp.636–658, <http://dx.doi.org/10.1108/IJESM-02-2021-0009>.
- Shmueli, G., Bruce, P.C., Gedek, P. and Patel, N.R. (2020) *Data Mining for Business Analytics: Concepts, Techniques, and Applications in Python*, John Wiley & Sons, NJ, USA.
- Wagenhofer, A. (2014) 'The role of revenue recognition in performance reporting', *Accounting and Business Research*, Vol. 44, No. 4, pp.349–379, <https://doi.org/10.1080/00014788.2014.897867>.
- Walmsley, T., Rose, A., John, R., Wei, D., Hlavka, J.P., Machado, J. and Byrd, K. (2023) 'Macroeconomic consequences of the COVID-19 pandemic', *Economic Modeling*, Vol. 120, pp.1–31, <https://doi.org/10.1016/j.econmod.2022.106147>.
- Wu, A. (2024) *The Story Behind Tesla's Success (TSLA)*, Investopedia [online] <https://www.investopedia.com/articles/personal-finance/061915/story-behind-teslas-success.asp> (assessed 10 July 2024).
- Zhou, Z.H. (2012) *Ensemble Methods: Foundations and Algorithms*, CRC Press, Boca Raton, FL, USA.

Appendix

Table A1 Configuration of best time series models for M7 companies

<i>M7 company</i>	<i>Description of model configurations</i>		
	<i>Model 1: two-level model</i>	<i>Model 2: HW model</i>	<i>Model 3: ARIMA model</i>
Alphabet	Quadratic trend and quarterly seasonality regression with MA for regression residuals with window $k = 2$	ets(M, M, M): multiplicative error, multiplicative trend, and multiplicative seasonality, $\alpha = 0.2984$, $\beta = 0.0918$, $\gamma = 1e-04$	ARIMA (1, 1, 2) (2, 1, 2): non-seasonal option: (1, 1, 2) seasonal option: (2, 1, 2)
Amazon	Quadratic trend and quarterly seasonality regression with MA for regression residuals with window $k = 2$	ets(M, A, M): multiplicative error, additive (A) trend, and multiplicative (M) seasonality, $\alpha = 0.5247$, $\beta = 0.1949$, $\gamma = 0.4753$	ARIMA (2, 1, 2) (2, 1, 1): non-seasonal option: (2, 1, 2) seasonal option: (2, 1, 1)
Apple	Quadratic trend and quarterly seasonality regression with MA for regression residuals with window $k = 2$	ets(M, M, M): multiplicative error, multiplicative trend, and multiplicative seasonality, $\alpha = 0.5506$, $\beta = 1e-04$, $\gamma = 0.3869$, $\phi = 0.8732$	ARIMA (2, 1, 2) (2, 2, 3): non-seasonal option: (2, 1, 2) seasonal option: (2, 2, 3)
Meta	Quadratic trend and quarterly seasonality regression with MA for regression residuals with window $k = 2$	ets(M, A, M): multiplicative error, additive trend, and multiplicative seasonality, $\alpha = 0.9773$, $\beta = 0.2965$, $\gamma = 0.0227$	ARIMA (2, 1, 2) (2, 2, 2): non-seasonal option: (2, 1, 2) seasonal option: (2, 2, 2)
Microsoft	Quadratic trend and quarterly seasonality regression with MA for regression residuals with window $k = 2$	ets(A, A, A): additive error, additive trend, and additive seasonality, $\alpha = 0.7217$, $\beta = 1e-04$, $\gamma = 1e-04$	ARIMA (2, 1, 2) (2, 1, 1): non-seasonal option: (2, 1, 2) seasonal option: (2, 1, 1)
NVIDIA	Quadratic trend and quarterly seasonality regression with MA for regression residuals with window $k = 2$	ets(M, A, M): multiplicative error, additive trend, and multiplicative seasonality, $\alpha = 0.9967$, $\beta = 1e-04$, $\gamma = 0.0033$	ARIMA (5, 1, 3) (5, 1, 2): non-seasonal option: (5, 1, 3) seasonal option: (5, 1, 2)
Tesla	Quadratic trend and quarterly seasonality regression with MA for regression residuals with window $k = 2$	ets(M, A, N): multiplicative error, additive trend, and no seasonality, $\alpha = 0.7582$, $\beta = 0.0616$, $\gamma = 0$	ARIMA (1, 1, 1) (1, 1, 1): non-seasonal option: (1, 1, 1) seasonal option: (1, 1, 1)